Computational Approaches using Machine Learning to Predict Functional Impact of SERPINA1 Missense Mutations

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Research Question

Can we predict pathogenicity in variants associated with SERPINA1 and AAT?

Introduction and Background

Alpha-1 antitrypsin (AAT) is a protease inhibitor made in the liver with the goal of protecting your lungs from irritants. The lack of AAT in the lungs may make them more easily damaged by smoking or through pollution and dust in the air [4]. An AAT deficiency is caused by mutations in the *SERPINA1* gene, the gene that carries the instruction for making the AAT protein [6].

Methods and Materials

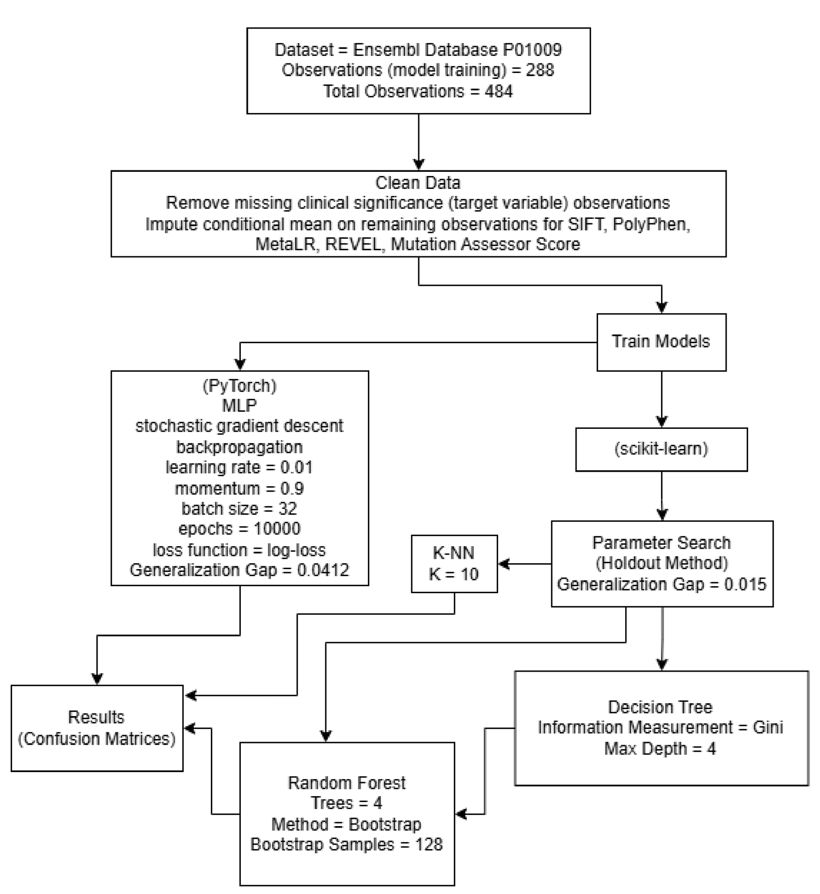

Figure 1:Methodology [11 - 18]

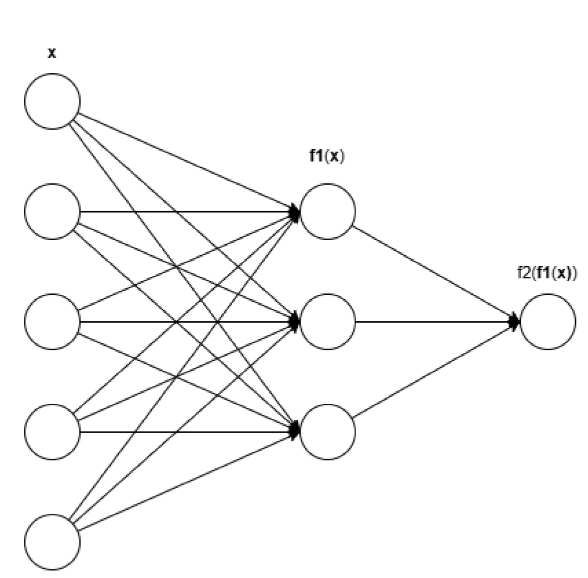

Figure 2:Model - MLP

$$\mathbf{ReLU}_i(\mathbf{x}) = \max\{0, x_i\}$$

$$\sigma(x) = \frac{1}{1+e^{-x}}$$

$$\mathbf{f}_1(\mathbf{x}) = \mathbf{ReLU}\left(\mathbf{x}\begin{bmatrix} -1.3321 & -4.2124 & -1.9546 \\ -4.2729 & 6.1888 & -2.3244 \\ 1.3101 & -5.3542 & 2.1324 \\ 0.3361 & 1.8541 & 0.0944 \\ -1.4612 & 1.2690 & -2.7183 \end{bmatrix} + \begin{bmatrix} 3.6913 \\ -0.0730 \\ 3.9457 \end{bmatrix}^T\right)$$

$$f_2(\mathbf{x}) = \left[\left[\sigma(\mathbf{x}\begin{bmatrix} -6.0766 \\ -9.3592 \\ -6.0285 \end{bmatrix} + 5.3688)\right]\right]$$

$$L(\mathbf{y}, \hat{\mathbf{y}}) = -\frac{1}{N}\sum y_i \ln(\hat{y}_i) + (1-y_i)\ln(1-\hat{y}_i)$$

$$W = W - \eta\nabla L(\mathbf{y}, \hat{\mathbf{y}}) + \alpha\Delta W$$

$$G(p_1, p_2) = 1 - p_1^2 - p_2^2$$

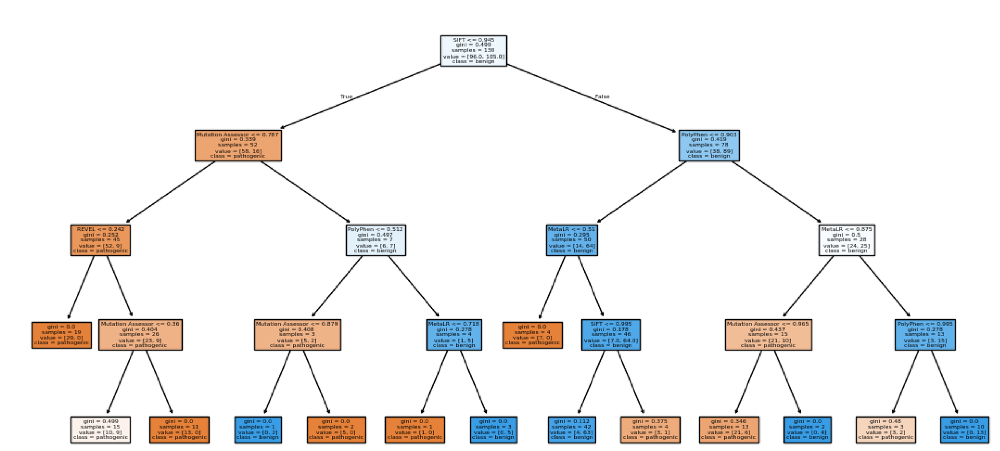

Figure 3:Model - Random Forest Tree 1

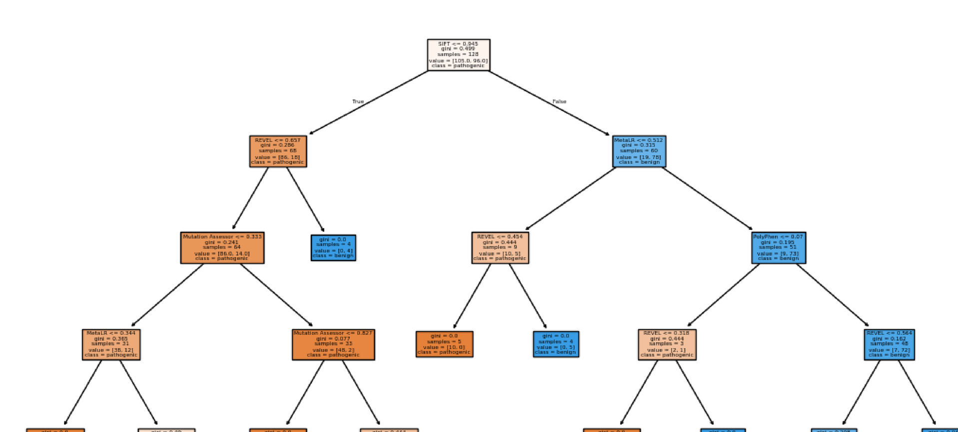

Figure 4:Model - Random Forest Tree 2

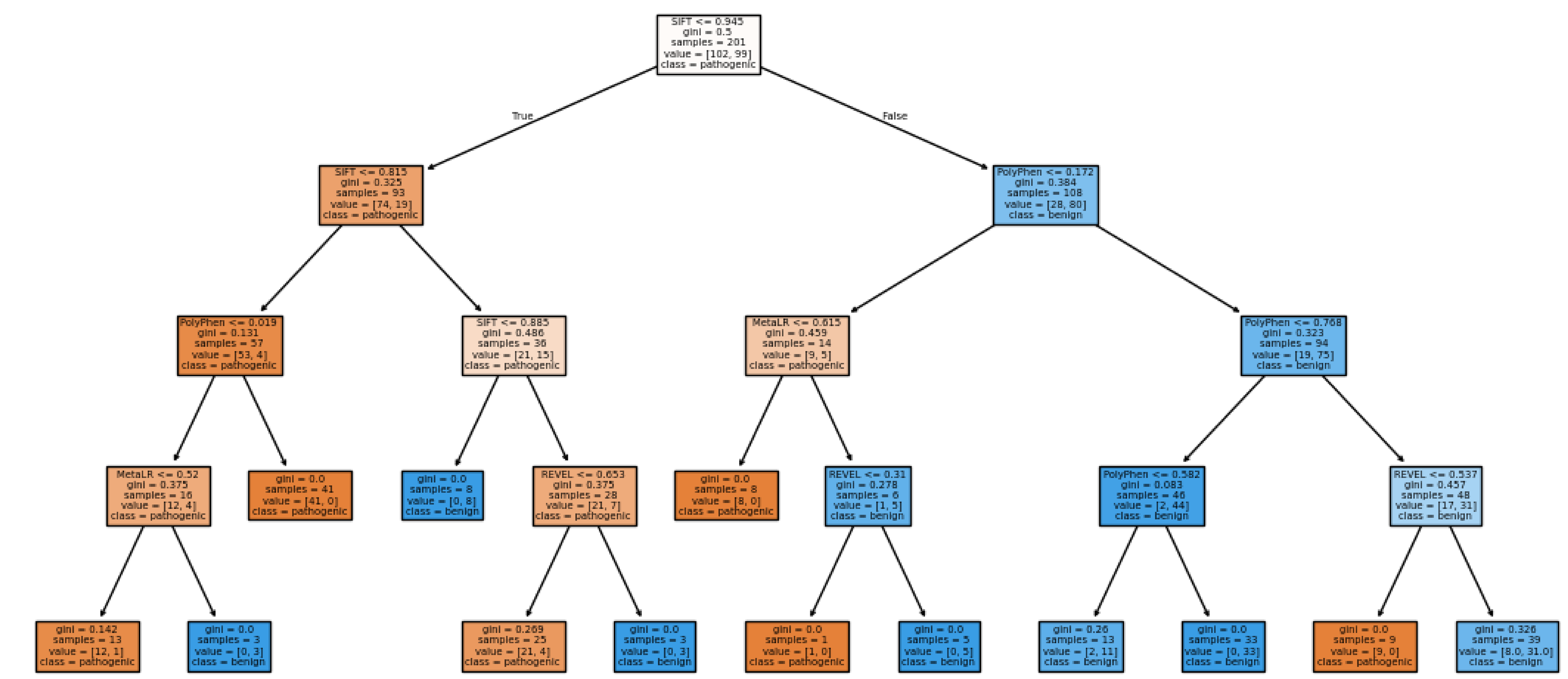

Figure 5: Model - Decision Tree

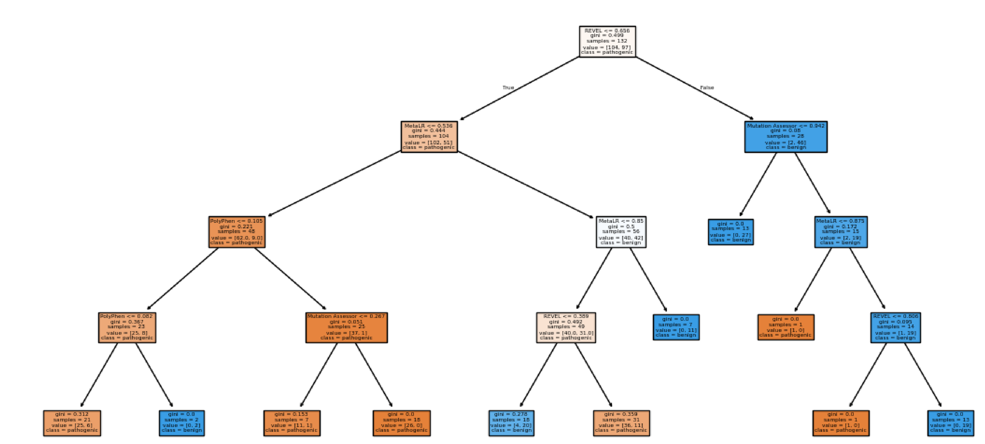

Figure 6:Model - Random Forest Tree 3

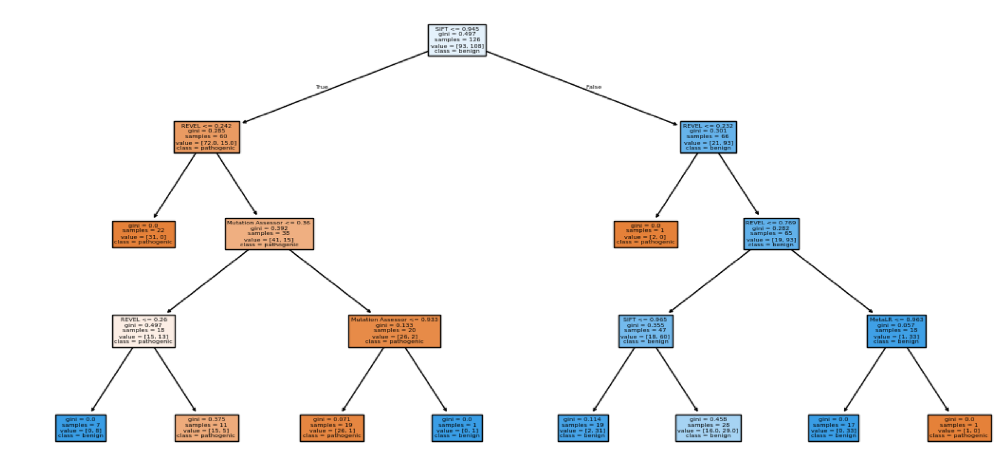

Figure 7:Model - Random Forest Tree 4

Results

The confusion matrices for each model are shown throughout this section. These four models resulted in an accuracy of determining whether any given VUS was benign or pathogenic of 86%, 89%, 90%, and 85% respectively.

MLP	Predicted Benign	Predicted Pathogenic
Benign	316	24
Pathogenic	42	102

Decision Tree	Predicted Benign	Predicted Pathogenic
Benign	293	47
Pathogenic	6	138

Random Forest	Predicted Benign	Predicted Pathogenic
Benign	299	41
Pathogenic	6	138

K-NN	Predicted Benign	Predicted Pathogenic
Benign	276	64
Pathogenic	7	137

Discussion

This study used bioinformatics and machine-learning approaches to investigate the pathogenic potential of missense variants in the SERPINA1 gene associated with alpha-1 antitrypsin (AAT) deficiency. From this, we conjecture the reason the Neural Network is able to predict benign better than pathogenic (opposed to all other models) to be caused from the ability of the neural network to capture more complex relationships. In the future, more models and additional activation functions can be experimented with to determine if more complex relationships exist.

References